Daily Progress Report (3rd June 2026)

Project Title: Design and Physical Implementation of a 4-bit Carry Look-Ahead Adder using CMOS Mirror Logic

Date: 03 June 2026

Intern Name: Akshathaa G V

Objective

To initiate the transistor-level design of a 4-bit Carry Look-Ahead (CLA) Adder using CMOS Mirror Logic in Cadence Virtuoso.

Work Completed

During the first day of the project, the Carry Look-Ahead Adder architecture and carry-generation equations were studied to understand the design methodology and implementation flow.

The following transistor-level schematics were created in Cadence Virtuoso:

- 2-Input AND Gate
- 2-Input OR Gate
- Carry Generation Circuit C1
- Carry Generation Circuit C2
- Carry Generation Circuit C3
- Carry Generation Circuit C4
- Sum Generation Circuit

The transistor connections and logic implementation for each block were completed and verified at the schematic level. Individual circuit blocks were organized for future integration into the complete 4-bit CLA design.

Tools Used

- Cadence Virtuoso
- 90 nm CMOS Technology Library

Outcome

The fundamental schematic blocks required for implementing the 4-bit Carry Look-Ahead Adder were successfully designed. The project has progressed to the stage where symbol creation, hierarchical integration, and functional simulations can be carried out.

Challenges Encountered

- Understanding the carry-generation equations and translating them into transistor-level CMOS implementations.

- Managing increasing schematic complexity for higher-order carry circuits.

Plan for Next Working Day

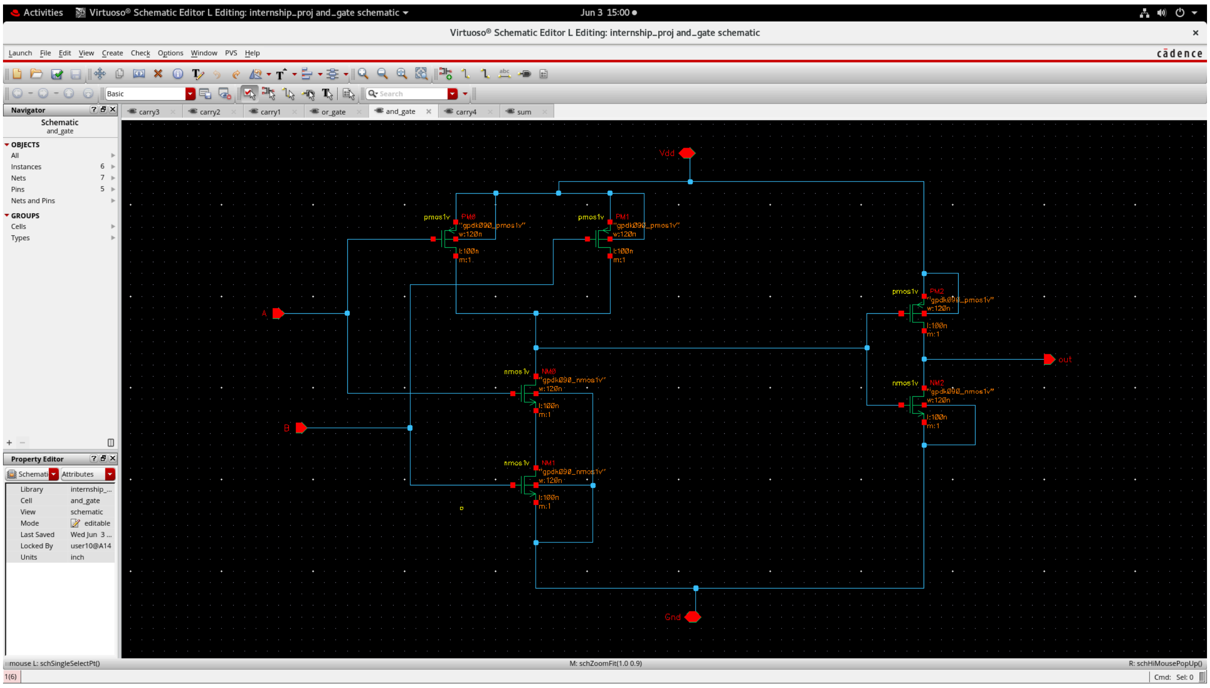
1. Create symbols for the completed schematic blocks.
2. Integrate the individual modules into a hierarchical 4-bit CLA structure.
3. Set up simulation environments in ADE.
4. Perform functional verification of carry and sum outputs.
5. Begin timing and performance analysis.

Remarks

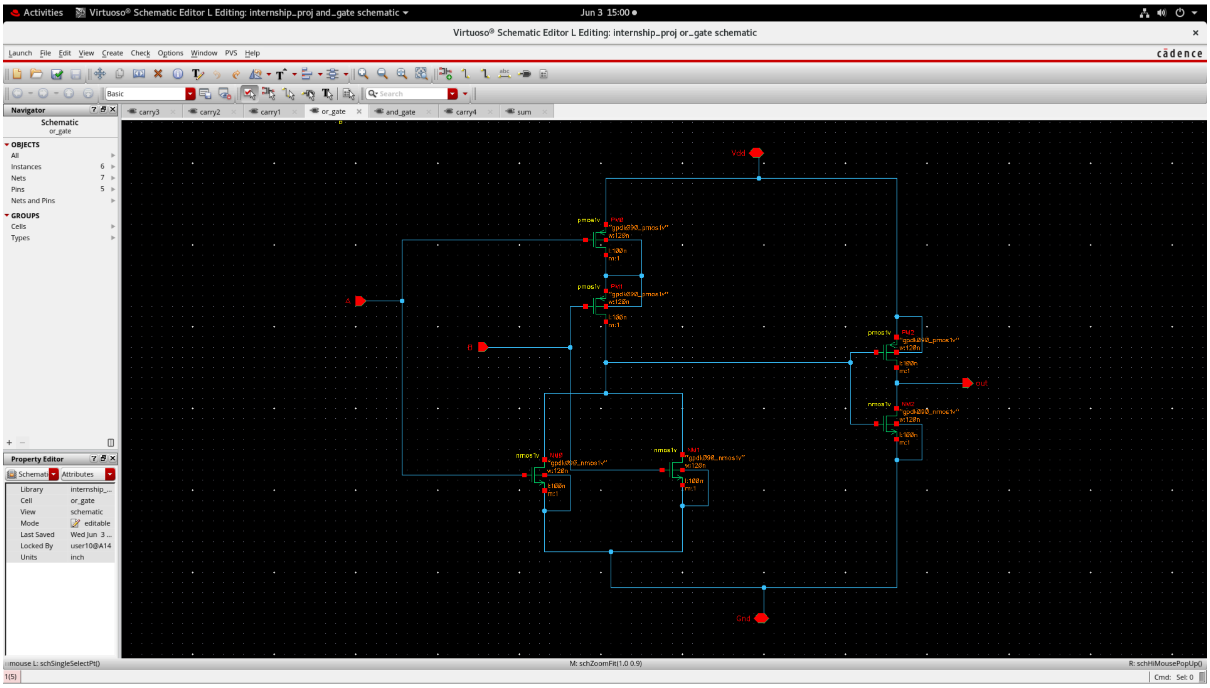
The initial schematic design phase has been completed successfully. The designed blocks provide the foundation for subsequent simulation, verification, and physical implementation stages of the project.

Schematic circuit

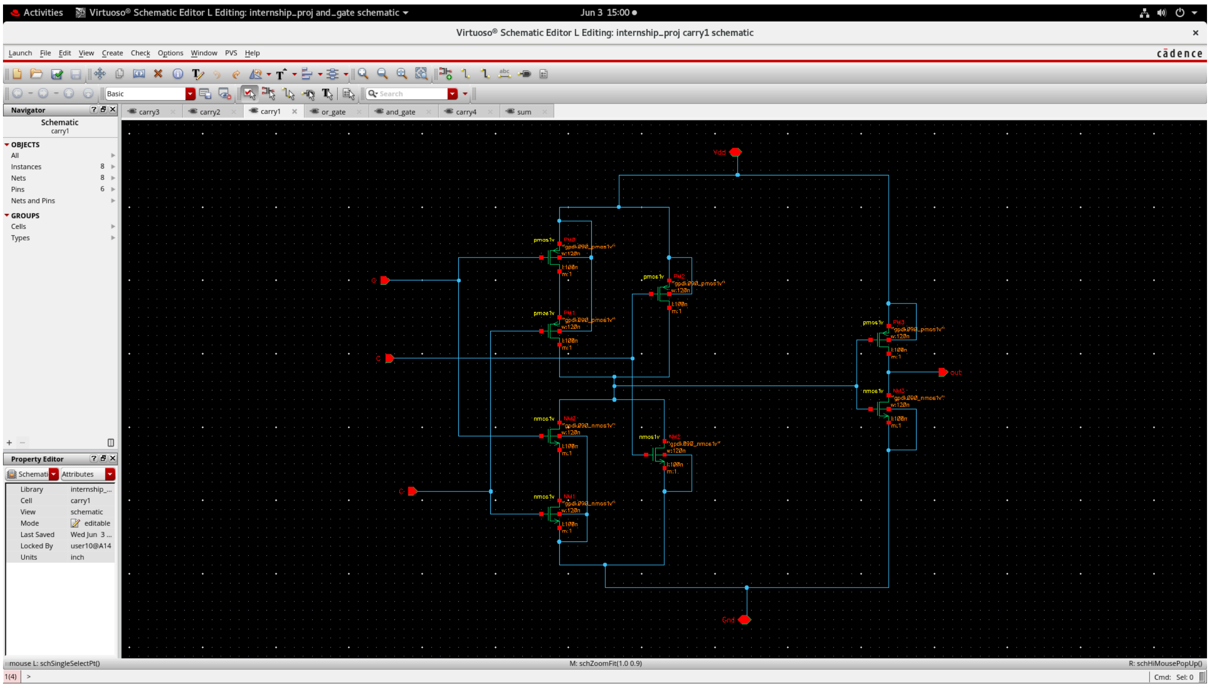
And gate 2 input:



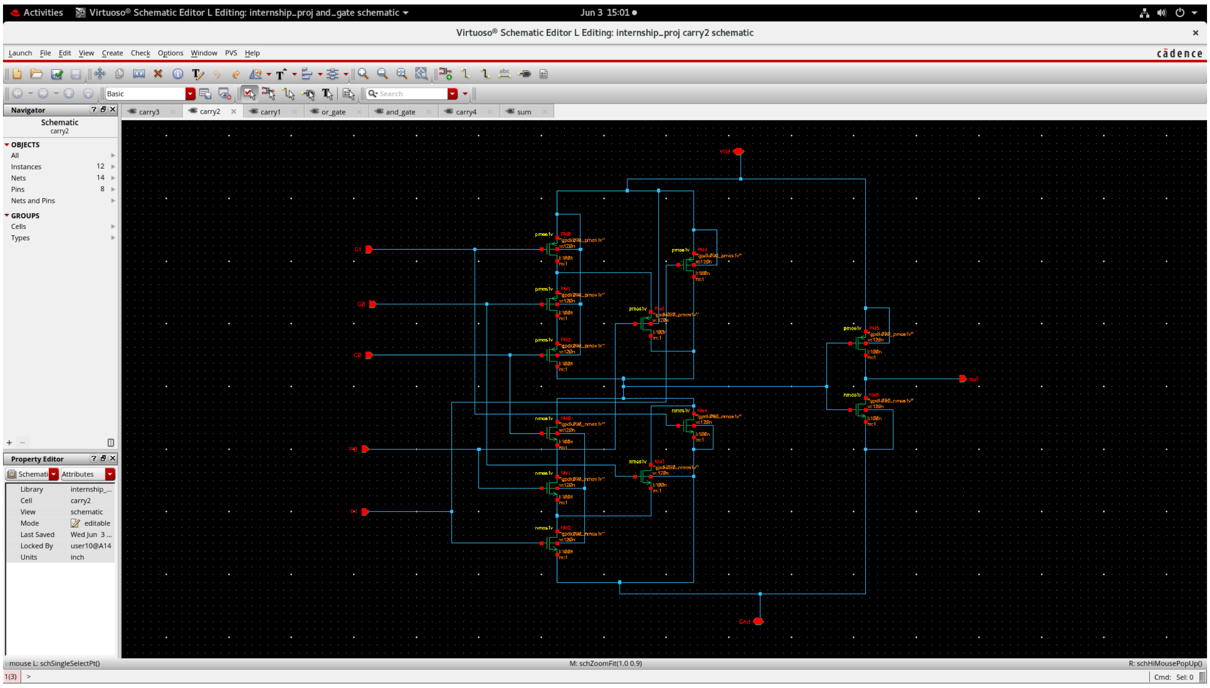
Or gate 2 input:



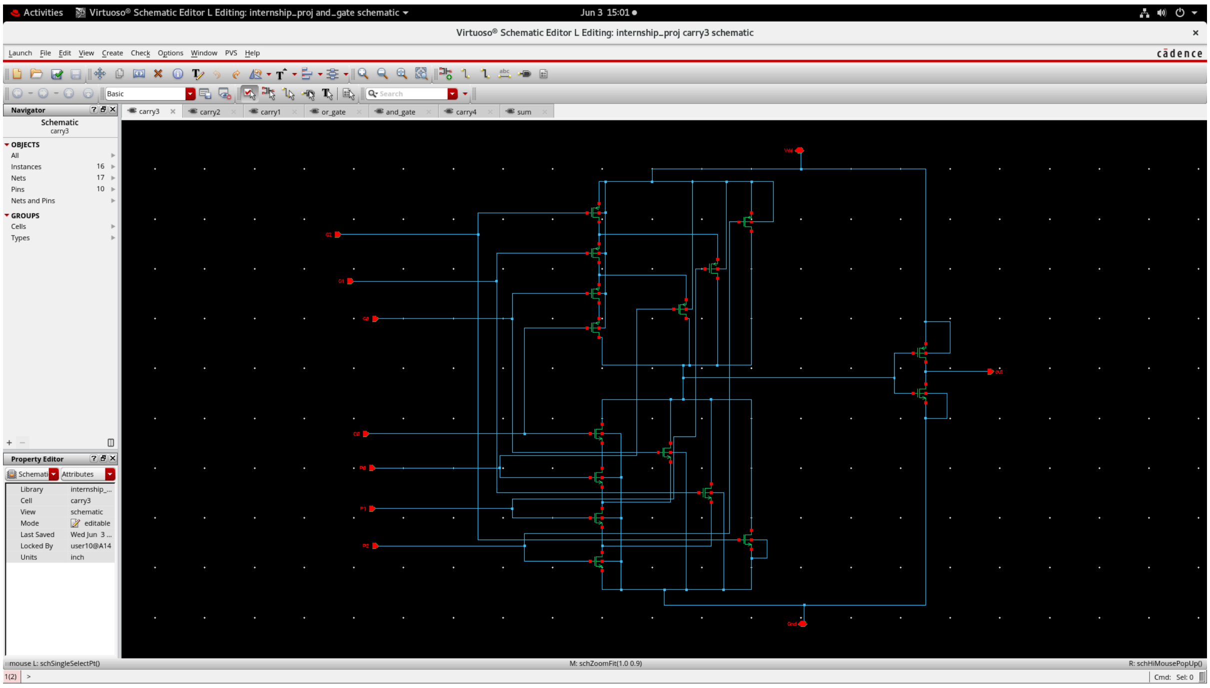
Carry 1:



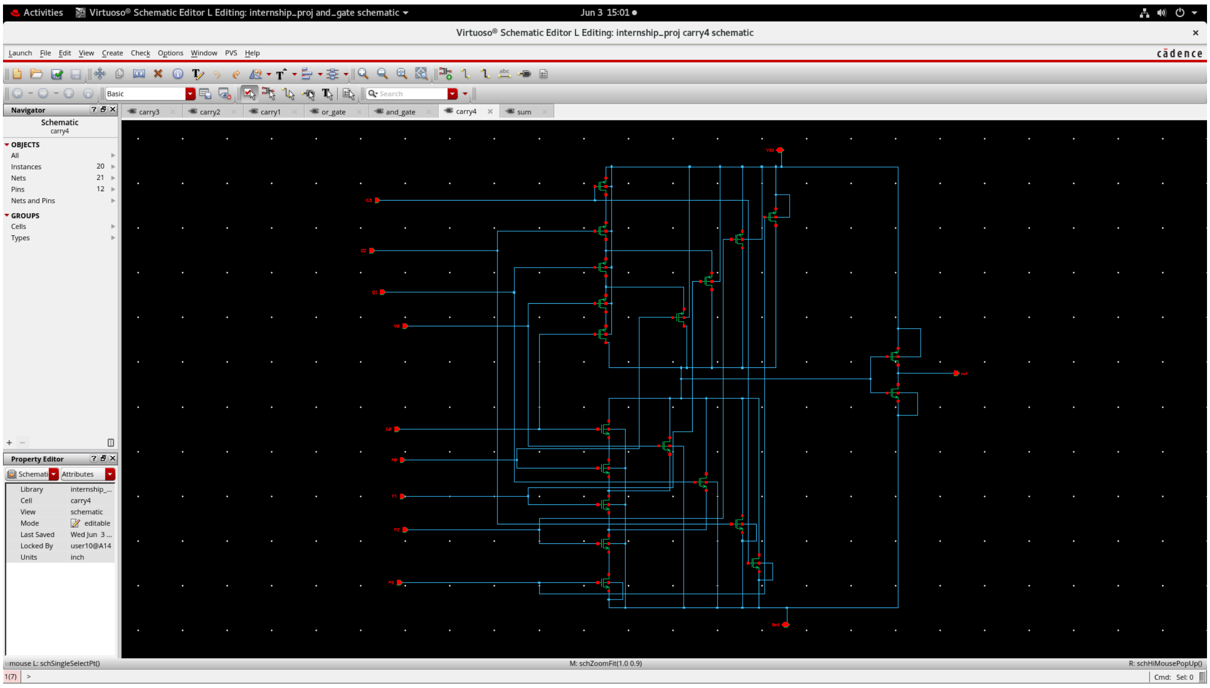
Carry 2:



Carry 3:



Carry 4: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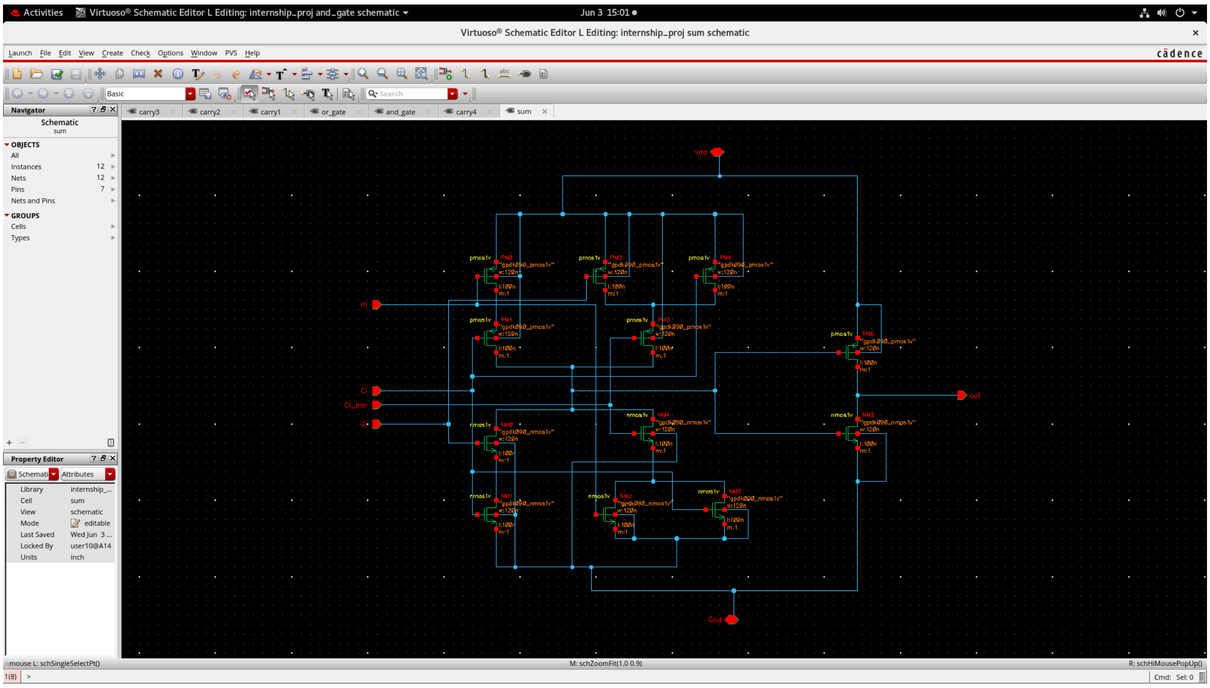
Sum: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